Multi-algorithmic approaches to mono and polyphonic pitch detection / estimation

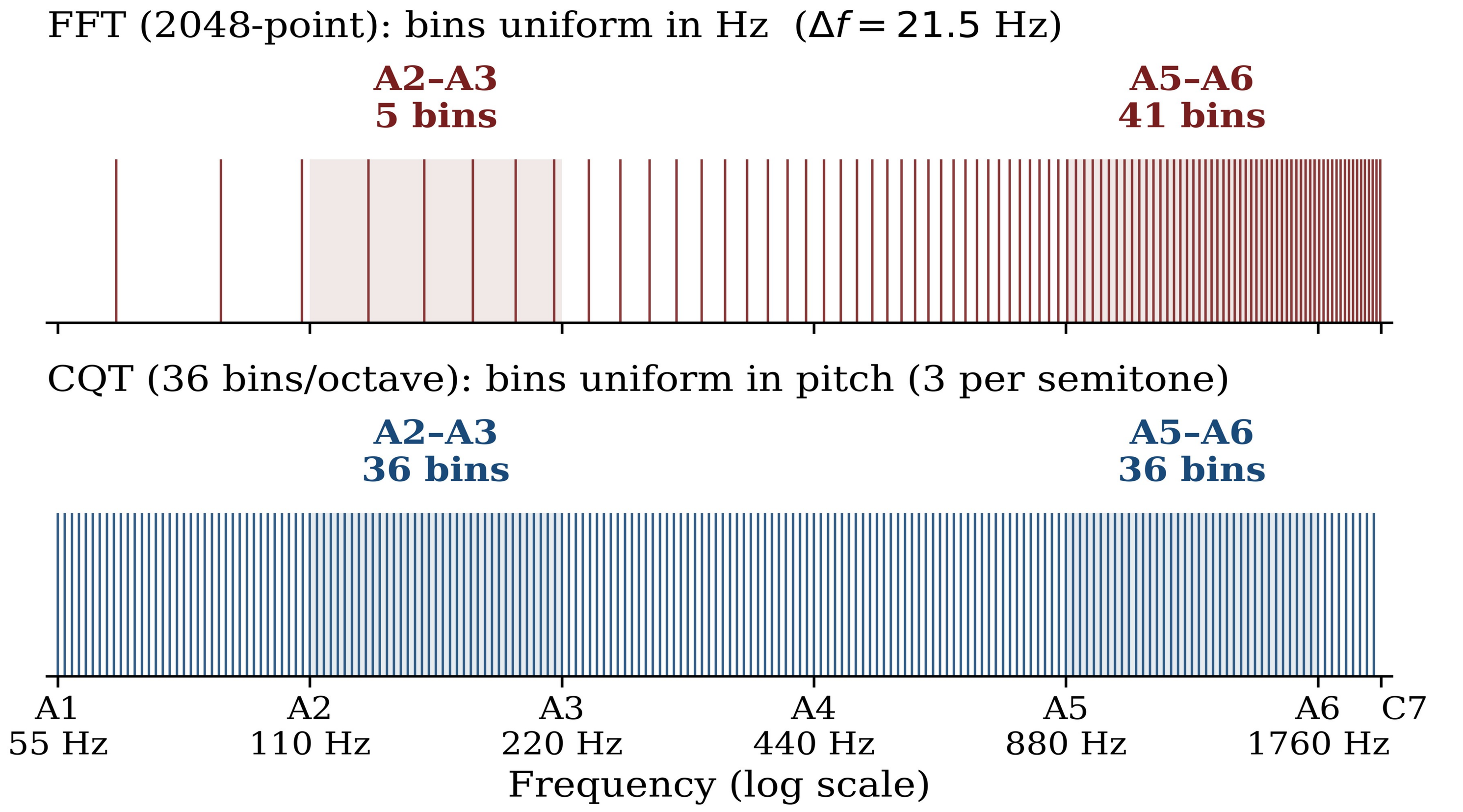
Joseph Lincoln, Dr. Mangolika Bhattacharya, Dr. Roy Magnuson | | Illinois State University Schools of Information Technology and Music | | Summer Research Symposium · July 2026

1.) Abstract

Real-time pitch detection is largely solved for single notes; our previous YIN-based **[1]** monophonic web tuner reaches ~95% accuracy, operating entirely client-side. Music, though, is polyphonic. P.I.T.C.H. (Polyphonic Interpretation Tool for Chords and Harmony) extends that work toward real-time chord and harmony labeling, modeling the physics and acoustics behind music theory before layering on any machine learning.

The interpretable core matches incoming sound against a hand-built library of harmonic templates (one per note), then reads chord quality and bass separately before voting on a label. Benchmarked on the Bach10 chorales against Spotify's Basic Pitch under an identical protocol, P.I.T.C.H. matches or beats that neural network on three of four instrument voices, and labels chords far more accurately (0.829) than note-by-note scoring suggests (0.665); evidence that harmony is a more robust target than pitch.

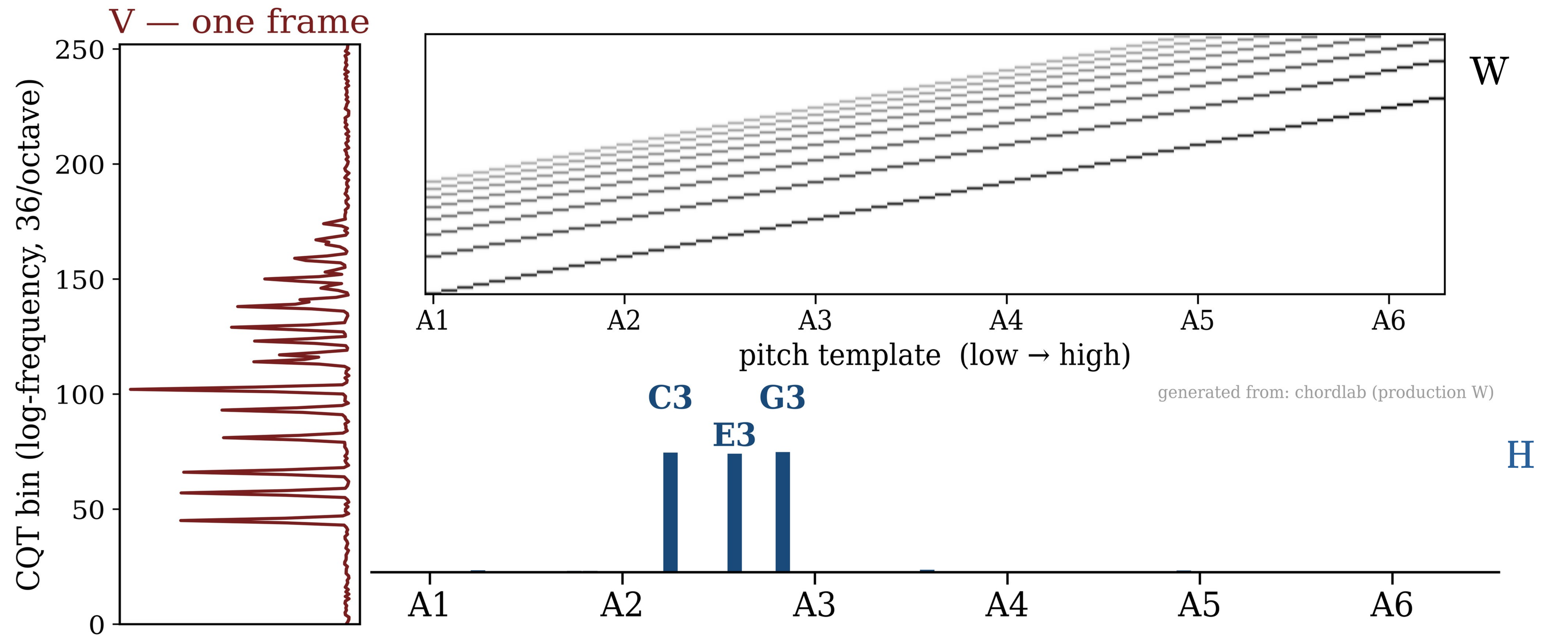
2.) Why constant-Q transforms (CQT) and not fast Fourier transforms (FFT)?



Linear FFT bin selection starves the bass register and floods the treble (or vice versa), which biases our results in the case of polyphonic interpretation, detection, and estimation. CQT, on the other hand, gives every semitone and octave range equal footing by evenly spacing out the bins.

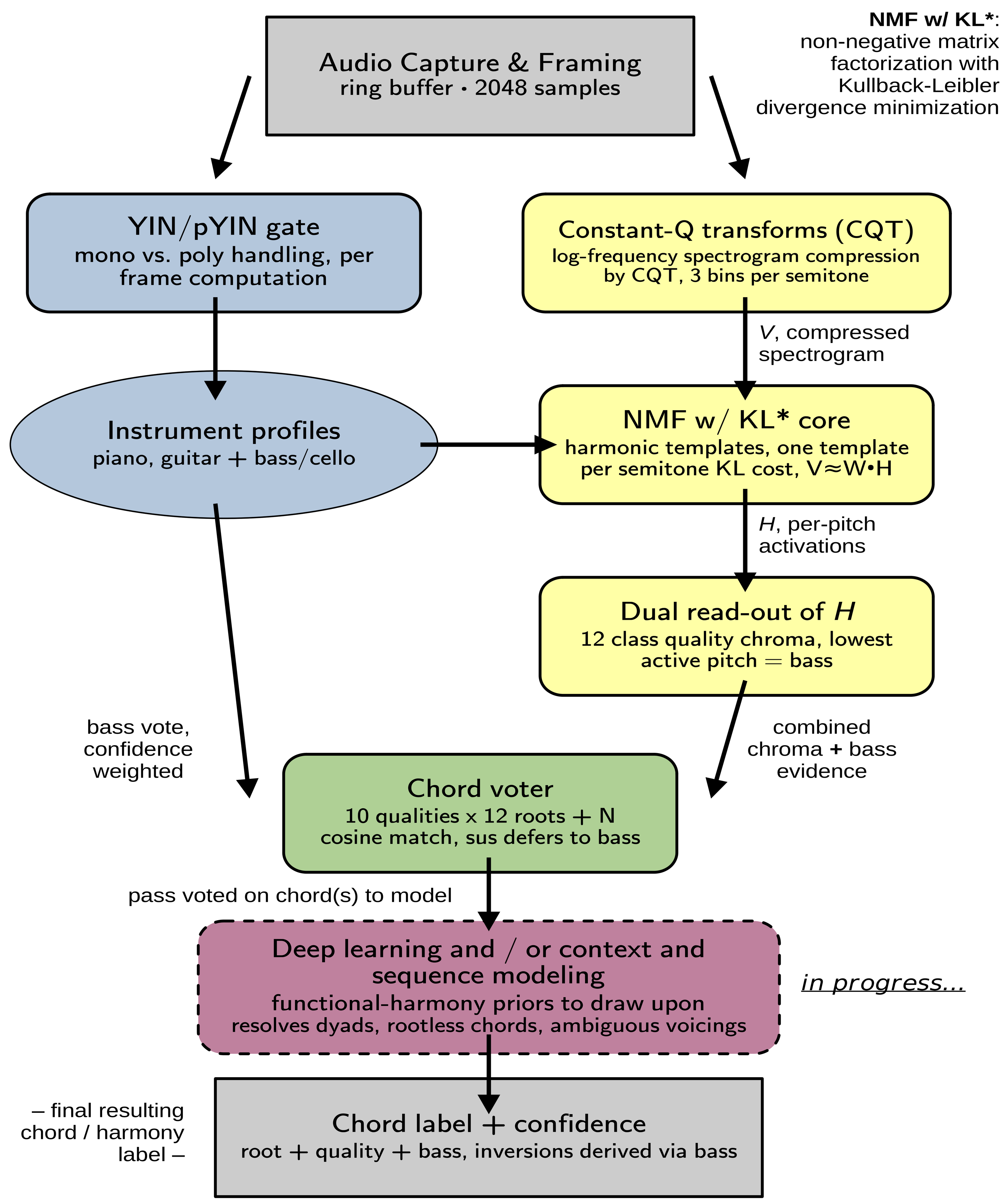
3.) Harmonic dictionary + NMF

Each frame is explained as $V \approx W \cdot H$; sparse activations H say which pitches are present (C3+E3+G3 → chroma {C, E, G}=major quality; lowest active pitch=bass → "C:maj")



One harmonic template per semitone, using NMF w/ Kullback-Leibler divergence*; each column of W is a single pitch's template, on a grid of 3 bins per semitone.

4.) P.I.T.C.H. v0.1 (Polyphonic Interpretation Tool for Chords and Harmony) Proof-of-Concept and Design Schema



5.) Evaluation protocol(s)

- **Dataset & split:** Bach10 **[3]**, ten different four-part chorales (violin, clarinet, saxophone, bassoon) with ground-truth pitch tracks. Pieces 01-03 are the development set, where every free parameter (thresholds, harmonic template dictionary choices, etc.) is determined by attuning to *only* these first three pieces; 04-10 are the proper, held-out test set, scored once.
- **Signal chain:** Instrument stems are summed to one uniform mix so every system analyzes identical audio; signal provenance is logged on every run.
- **Frame metric:** mir_eval **[4]** multi-pitch Accuracy and Chroma Accuracy on a 10 ms grid. Standard, but a *proxy*; it scores pitches, *not* harmony.
- **Chord metric:** Reference chord labels are produced by running the *same voter* on the ground-truth pitches. Any disagreement is measuring pure estimation loss, *not* disagreement about chord labeling conventions.
- **External baseline:** Spotify's Basic Pitch (CNN-based) evaluated under the *identical* protocol: same mixes, same grid, detection threshold swept on the same first three development pieces.

*Non-negative Matrix Factorization using Kullback-Leibler Divergence** ***[2]***

$$D(A \| B) = \sum_{ij}\left(A_{ij}\log\frac{A_{ij}}{B_{ij}} - A_{ij} + B_{ij}\right)$$

D measures how far a reconstruction, B, falls from an observation, A. P.I.T.C.H. minimizes D(V ‖ W·H), which is the mismatch between the observed spectrum and the template combination meant to explain it. Kullback-Leibler divergence is used, rather than squared error, because it weights *relative* mismatch, so quiet upper partials still count.

6.) Results: frame / proxy, per-voice, and task domain

Per-voice recall – quartet, Bach10 test set

	P.I.T.C.H. v0.1	Spotify's Basic Pitch
Clarinet	**0.90 – 0.99**	0.86 – 0.89
Saxophone	**0.90 – 0.99**	0.86 – 0.89
Bassoon	**0.90 – 0.99**	0.86 – 0.89
Violin	0.45 – 0.67	**0.76 – 0.87**

Interpretable model matches or beats Spotify's CNN-based Basic Pitch **[5]** model on three out of four voices. The violin deficit is mechanistic: soprano fundamentals coincide with lower-voice partials (G4 = fourth partial of G2), leaving shared CQT bins un-arbitrable within a single frame.

Frame-level accuracy – Bach10 test set (pieces 04-10, scored once)

	P.I.T.C.H. v0.1		Spotify's Basic Pitch	
	Acc	**ChromaAcc**	**Acc**	**ChromaAcc**
Duet (vln + bsn)	0.622	0.638	0.756	0.783
Trio	0.648	—	—	—
Quartet	**0.665**	0.686	**0.781**	0.803

Duet values include the adopted 5-frame mean temporal smoothing. Identical protocol for both systems: same mixtures, same 10 ms grid, thresholds swept on development pieces 01-03 only. Trio ensemble configurations not benchmarked against external baseline.

Chord-level agreement (P.I.T.C.H. vs. *reference*)

reference = same chord voter, but applied directly to provided Bach10 ground-truth pitches

	Exact	Root	Quality	Bass
Quartet	**0.829**	0.892	0.833	0.876
Duet	0.564	0.818	0.570	**0.936**

Chord- and harmony-level accuracy exceeds the frame-level proxy accuracy (0.829 vs. 0.665); dyads under-determine quality while bass and root survive, which jointly mandates the context and sequence modeling to be added in the next phases of the project.

References:

[1] de Cheveigné, A., & Kawahara, H. (2002). YIN, a fundamental frequency estimator for speech and music. *The Journal of the Acoustical Society of America, 111*(4), 1917-1930. https://doi.org/10.1121/1.1458024.

[2] Lee, D. D., & Seung, H. S. (2000). Algorithms for non-negative matrix factorization. *Advances in Neural Information Processing Systems, 13*, 556–562.

[3] Duan, Z., Pardo, B., & Zhang, C. (2010). Multiple fundamental frequency estimation by modeling spectral peaks and non-peak regions. *IEEE Transactions on Audio, Speech, and Language Processing, 18*(8), 2121–2133. https://doi.org/10.1109/TASL.2010.2042119.

[4] Raffel, C., McFee, B., Humphrey, E. J., Salamon, J., Nieto, O., Liang, D., & Ellis, D. P. W. (2014). mir_eval: A transparent implementation of common MIR metrics. In *Proceedings of the 15th International Society for Music Information Retrieval Conference (ISMIR)* (pp. 367–372).

[5] Spotify. (2023). Basic Pitch [Computer software]. GitHub. https://github.com/spotify/basic-pitch.

Acknowledgments — the authors gratefully acknowledge the Illinois State University Schools of Information Technology and Music for their support and funding of necessary materials and/or printing costs, and extend a thank you to the faculty mentors overseeing this project.